

tf.eye Stay organized with collections

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https://www.tensorflow.org/api_docs/python/tf/eye

Construct an identity matrix, or a batch of matrices.

Function Signatures

```
tf.eye( num_rows, num_columns=None, batch_shape=None,  
dtype=tf.dtypes.float32, name=None )
```

Code Examples

```
tf.eye(  
    num_rows,  
    num_columns=None,  
    batch_shape=None,  
    dtype=tf.dtypes.float32,  
    name=None  
)
```

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num_rows,  
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```

```
tf.dtypes.float32
```

```
# Construct one identity matrix.  
tf.eye(2)  
==> [[1., 0.],  
[0., 1.]]  
# Construct a batch of 3 identity matrices, each 2 x 2.  
# batch_identity[i, :, :] is a 2 x 2 identity matrix, i = 0, 1, 2.  
batch_identity = tf.eye(2, batch_shape=[3])  
# Construct one 2 x 3 "identity" matrix  
tf.eye(2, num_columns=3)  
==> [[ 1.,  0.,  0.],  
[ 0.,  1.,  0.]]
```

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# Construct one identity matrix.  
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# Construct one 2 x 3 "identity" matrix  
tf.eye(2, num_columns=3)  
==> [[ 1.,  0.,  0.],  
[ 0.,  1.,  0.]]
```

Documentation

Construct an identity matrix, or a batch of matrices.

View aliases

Main aliases

`tf.linalg.eye`

See

Migration guide for
more details.

`tf.compat.v1.eye`, `tf.compat.v1.linalg.eye`

Used in the notebooks

- Advanced automatic differentiation
 - Introduction to graphs and `tf.function`
 - TensorFlow Probability Case Study: Covariance Estimation
 - Learnable Distributions Zoo
 - A Tour of TensorFlow Probability
 - Bayesian Gaussian Mixture Model and Hamiltonian MCMC
 - Bayesian Modeling with Joint Distribution

See also `tf.ones`, `tf.zeros`, `tf.fill`, `tf.one_hot`.

Returns

